SRCBTFusion-Net: An Efficient Fusion Architecture via Stacked Residual Convolution Blocks and Transformer for Remote Sensing Image Semantic Segmentation

Junsong Chen, Jizheng Yi, *Member, IEEE*, Aibin Chen, and Hui Lin

***Abstract*— Convolutional neural network (CNN) and transformer-based self-attention models have their advantages in extracting local information and global semantic information, and it is a trend to design a model combining stacked residual convolution blocks (SRCBs) and transformer. How to efficiently integrate the two mechanisms to improve the segmentation effect of remote sensing (RS) images is an urgent problem to be solved. An efficient fusion via SRCB and transformer (SRCBTFusion-Net) is proposed as a new semantic segmentation architecture for RS images. The SRCBTFusion-Net adopts an encoder–decoder structure and the Transformer is embedded into SRCB to form a double coding structure, then the coding features are upsampled and fused with multiscale features of SRCB to form a decoding structure. First, a semantic information enhancement module (SIEM) is proposed to get global clues for enhancing deep semantic information. Subsequently, the relationship guidance module (RGM) is incorporated to reencode the decoder's upsampled feature maps, enhancing the edge segmentation performance. Second, a multipath atrous self-attention module (MASM) is developed to enhance the effective selection and weighting of low-level features, effectively reducing the potential confusion introduced by the skip connections between low- and high-level features. Finally, a multiscale feature aggregation module (MFAM) is developed to enhance the extraction of semantic and contextual information, thus alleviating the loss of image feature information and improving the ability to identify similar categories. The proposed SRCBTFusion-Net's performance on the Vaihingen and Potsdam datasets is superior to the state-of-the-art methods. The code will be freely available at https://github.com/js257/SRCBTFusion-Net.**

***Index Terms*— Multiscale feature, remote sensing (RS), semantic segmentation, transformer.**

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I. Introduction

SEGMENTATION of remote sensing (RS) images is an essential stage in the process of RS image processing, and the subsequent RS image processing will be significantly impacted by the segmentation quality. Semantic segmentation, as a feasible solution, has been widely discussed. Image semantic segmentation provides category information, position information, attitude information, and environment information about objects in images, which clearly and intuitively present the rich details of ground objects and the relationship between adjacent ground objects. As an important branch in the field of semantic segmentation, semantic segmentation of RS images automatically extracts the surface features in RS images and assigns semantic categories to ground objects, which has advanced significantly in many fields, such as environmental monitoring [1], urban planning [2], and disaster warning [3]. However, the RS image contains complex information about ground objects, which makes this research challenging.

A. Lack of Sufficient Global Context Information and Fine Spatial Features

The RS images usually contain boundary crossings, building corners, and small-scale ground objects. In the existing methods, the low-order fine-grained detailed feature mapping generated by the encoder is coupled with the high-order coarse-grained semantic information generated by the decoder without any additional refining resulting in insufficient fusion of features and absence of discrimination. One solution is to adopt a pyramid pool module [4] or context coding module [5] to aggregate multiscale context information. Another one is to adopt an attention mechanism [6] to solve. Although the context gathered by the aforementioned strategies is conducive to representing objects of various sizes, it does not consider the context dependencies of different categories and does not make full use of the long-range correlation of feature graphs, thus limiting the ability to deal with complex scenes.

B. Edge Segmentation Blur

Recently, many researchers have introduced boundary detection strategies to further distinguish adjacent semantic information. He et al. [7] embedded the proposed boundary enhancement module between the encoder and decoder to

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learn boundary features, then fused the acquired boundary features and semantic features and delivered them to the decoder to produce the prediction map. Diakogiannis et al. [8] introduced the Tanimoto loss function based on dice, which brought a balanced gradient to all tasks during training and improved the segmentation effect. The boundary-aware feature propagation (BFP) model [9] adopted one-way acyclic graph branches to optimize edges. Although the edge could be optimized by introducing the above strategy, it is still challenging to get a distinctive edge feature. This is because boundary pixels usually contain mixed features of multiple pixels and complex factors such as shadows.

The above approaches based on convolutional neural network (CNN) have their defects (locality and translation invariance), which restrict the model's capacity to identify long-range relationships and encode global context information. Since the transformer can establish associations between different locations and capture long-distance dependencies in sequences, the scheme of embedding the Transformer into CNN has become popular. Chen et al. [10] proposed the serial structure of CNN and transformer as the encoder, and the self-concerned feature is upsampled and combined with the multiscale CNN feature obtained from the coding path by skipping connection as the decoder for medical image segmentation. Inspired by this network, He et al. [11] improved the encoder and designed a parallel double encoding structure where UNet [12] is the primary encoder and Swin transformer is the auxiliary encoder. The segmentation effect of their model applied to the field of high-resolution RS images is better than many pure CNN models, which shows that the combination of CNN and Transformer makes up for the defects of the CNN model.

Given the above analysis, an efficient fusion via stacked residual convolution blocks (SRCBs) and transformer (SRCBTFusion-Net) is proposed for RS image segmentation. The SRCBTFusion-Net adopts the double coding structure of serial connection of SRCB (the first four layers of classic CNN model Resnet50 [13] with pretraining weights) and transformer, and the decoder is composed of the sequential upsampling of encoder output and the fusion of different resolution SRCB layers with skip connection. The following is a summary of the significant contributions.

1) The proposed semantic information enhancement module (SIEM) extracts global clues from the channel and spatial level to guide the decoder, and then it combines with the relationship guidance module (RGM) to decrease the absence of detailed information and enhance the discrimination between edge features and nonedge pixel features.
2) The multipath atrous self-attention module (MASM) combined with stacked dilated convolution blocks and an improved self-attention structure is proposed to enhance the effective selection and weighting of low-level features. This approach improves the grasp of both local details and global semantics, effectively reducing the potential confusion introduced by the skip connections between fused low- and high-level features.
3) A multiscale feature aggregation module (MFAM) is constructed for fusing multiscale features, which enhances the extraction of semantic and contextual information, alleviates the loss of image feature information and improves the ability to identify similar categories. To compensate for the lack of detailed features in the upsampling process, a detail compensation module (DCM) is proposed in MFAM to extract channel-related information, which improves the fine-grained feature representation capabilities.
4) Compared with the state-of-the-art approaches, our SRCBTFusion-Net achieves the optimum segmentation accuracy on two datasets, and the superiority of SRCBTFusion-Net is proved by a large number of visual analyses.

The remainder of this article is arranged as follows. Section II reviews the related research on semantic segmentation. Section III outlines in detail the proposed SRCBTFusion-Net structure. Section IV describes the experimental procedure, results, and visualization effects. Section V concludes and hints at the direction of future research.

II. Related Work

A. Review the Related Research on Semantic Segmentation

The semantic segmentation tasks of optical RS images are to label the optical RS images with semantic tags, classify the pixels belonging to the same category in the images into one category, and obtain the segmentation results. Unlike scene classification [14], which focuses on macro information, semantic segmentation is refined to each pixel. According to the fineness and method of the segmentation object, semantic segmentation can be separated into region-based and pixel-based segmentation methods.

Region-based segmentation methods usually divide optical RS images into regions according to certain strategies to obtain multiple image blocks, and then identify the category information of each image block and label it with semantic labels to obtain segmentation results. Region-based segmentation methods mainly include optical RS image segmentation methods based on a threshold, clustering, superpixel, graph theory, and Markov random field. Naik et al. [15] improved the segmentation effect by a multilevel threshold based on context-sensitive entropy and introduced an unbalanced optimizer to lower the calculation cost of the model. Trias-Sanz et al. [16] adopted a hierarchical clustering method to segment targets by transforming the original color space of optical RS images and combining texture and other information. Hadavand et al. [17] first divided the optical RS image into multiple regions by the simple linear iterative clustering (SLIC) superpixel algorithm, then clustered the regions by the improved density-based spatial clustering of application with noise (DBSCAN) algorithm, and further divided the noncore regions in the core segment by the Otsu threshold method. Finally, the land cover map was generated by a random forest classifier. Bao and Yao [18] roughly segmented the optical RS image by the quadtree segmentation method, then fused the spectrum, shape, texture, and other features of the optical RS image, and calculated the similarity by a multifeature algorithm based on graph theory to get the final segmentation result. Region-based segmentation methods can achieve the

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effect of segmentation, but these methods such as threshold, clustering, superpixel, graph theory, and Markov random field have their emphasis. For example, the threshold segmentation method is suitable for images with a strong contrast between the target and the background, and it has excellent advantages in processing images with extensive data because of its simple calculation, but it is not practical for images with noise and similar gray levels between the target and the background. The clustering algorithm has a good segmentation effect on images with noise points, but the calculation is complicated and unsuitable for images with extensive data. Moreover, these methods only rely on low-level visual features, which are not ideal for segmentation accuracy and speed, and it is difficult to achieve fine segmentation.

Pixel-based segmentation method can mark all pixels of an optical RS image with semantic labels, which can realize fine pixel-level segmentation. In recent years, CNN has made great progress in computer vision tasks because of its efficient feature learning ability. For example, a full convolution network (FCN) [19] has been frequently employed in semantic segmentation tasks. However, ordinary FCN can not restore the resolution of the input image, and it does not perform well in segmenting small details. In the task of RS image segmentation, multiscale spatial context information plays an important role in improving segmentation accuracy. Although ordinary convolution operations can capture a certain range of context information, it is difficult to capture multiscale spatial context information because of the limited receptive field of ordinary convolution. To solve this problem, the encoder–decoder structure is proposed and shows excellent performance in semantic segmentation tasks. For example, UNet [12] aggregated semantic information through feature stacking and fused features at both ends of the codec through a simple skip connection. The SegNet [20] adopted the maximum pooled position index to assist the decoder in recovering the resolution of the input image by upsampling, which reduced the model parameters and optimized the segmentation results. Although the network as mentioned above has made significant progress in semantic segmentation, when interpreting large-scale high-resolution RS images with complex backgrounds, it ignores these characteristics, such as large scale, a large amount of information, small targets, and blurred boundaries, which leads to the problems of poor segmentation results, low accuracy, and slow segmentation speed when interpreting RS images.

In order to overcome the above problems, many improved networks based on spatial pyramid pooling (SPP) have emerged. The SPP of PSPNet [21] aggregated context information through four pools of different sizes, which improved the problem of poor segmentation of small-scale targets and similar targets. Deeplabv3+ [22] is one of the most successful networks in the field of semantic segmentation. It introduced the atrous SPP (ASPP) to obtain multiple receptive fields, which could encode multiscale environmental information, and it optimized the target edge in the decoder to improve the segmentation performance. In addition, in order to introduce an attention mechanism into the semantic segmentation of RS images, researchers strive to combine the attention mechanism with other theories or algorithms. Ding et al. [23] proposed a method with both a block attention module and an attention embedding module, which enhanced the embedding of context information and enriched the semantic information of the underlying features. To further optimize the segmentation results and distinguish nearby similar targets, Chen et al. [24] proposed a boundary-enhanced attention module (BEAM) based on deep semantic information, which was combined with the previous position attention module (PAM) [6] to be superimposed with the original feature map to obtain the enhanced semantic boundary effectively. Cong et al. [25] proposed a mirror-symmetric boundary guidance (MBG) module, which introduced low-level boundary information to optimize and guide the boundary in the feature decoding stage, and adopted explicit supervision between the generated boundary map and the boundary ground truth value acquired by Canny algorithm, which effectively improved the accuracy of boundary prediction.

B. Review the Semantic Segmentation Methods Based on CNN and Transformer

In recent years, the model based on the transformer has obtained excellent results in many computer vision and natural language processing (NLP) tasks. The transformer adopts the self-attention layer to solve the problems of the limited receptive field and inadaptability to input content based on the CNN model. The existing RS image segmentation methods based on transformer usually adopt the mixed design, aiming at combining the advantages of CNN in spatial position representation and transformer in global relationship modeling. Wang et al. [26] adopted a patch segmentation module to divide the input image into nonoverlapping patches as "marks" and sent them to the encoder composed of cascaded Swin transformer to obtain multistage features. In the decoding stage, the spatial attention and channel attention composed of convolution layers were adopted to fuse the multistage features and then restored to the primary image scale for prediction. To tackle the issue that the original transformer can not deal with the similarity and computational complexity between hyperspectral images, Cai et al. [27] proposed a pure transformer model similar to the UNet structure. Among them, the spectral-wise multihead self-attention (S-MSA) was presented to capture the dependence between long-range spectra, and the mask-guided mechanism (MM) was employed to guide the S-MSA module to focus on areas with high-fidelity spectral representation. However, using a transformer as an encoder will lead to the scarcity of local features, which is inconvenient for segmenting edge pixels and small-scale objects. Therefore, many methods adopt CNN as an encoder and transformer as a decoder to solve this problem [28], [29].

Currently, the most widely adopted is the dual coding structure of CNN and transformer. The depth feature generated by the dual encoders is delivered to the decoder, and then gradually upsampled and restored to the primary image scale for segmentation. For example, in light of the absence of foreground saliency modeling in current RS images, Xu et al. [30] embedded a transformer into CNN to adaptively decrease background noise and increase object saliency by combining multiscale characteristics. In addition, a saliency-guided loss guidance network was proposed to focus on foreground saliency response. Zhang et al. [31] proposed a strategy that

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the local feature pyramid generated by CNN was input into the semantic extractor based on vision transformer (ViT) to encode the global semantics, and then fused the local multiscale features in the decoder stage to get the prediction label with enhanced representation. Different from the above method, to overcome the issue of structural learning loss induced by entering patches into a transformer and compressing them into 1-D sequences, Gao et al. [32] input images into a parallel dual-coding structure, and then the output features of the two encoders were fused in turn from the deepest layer and then upsampled to the input resolution. Inspired by the above excellent works, SRCBTFusion-Net adopts a cascade double coding structure (SRCB as the primary encoder and Transformer as the auxiliary encoder) and then integrates multistage features to up-sample in turn to restore the resolution of the input image.

III. Methodology

A. Overall Architecture

The general design of the proposed SRCBTFusion-Net is presented in Fig. 1. The whole structure adopts the double coding structure of RCSN and transformer cascade and also draws lessons from the skip connection structure of UNet. We employ the former four layers of SRCBs of Resnet50 to encode the input image $X \in R^{C\times H\times W}$ as the advanced feature representation. In order to overcome the defects of SRCB in obtaining global context information, we cascade eight layers of transformer stacked blocks behind the SRCB layer to obtain the encoded feature $Q \in R^{(H\times W/P^2)\times D}$. In order to restore spatial order, it is necessary to reshape $(H \times W/P^2)$ to $(H/P) \times (W/P)$, and then adopts 3×3 convolution to adjust the channel to 512. Then, the feature map is upsampled to $(H/8) \times (W/8)$ and $(H/4) \times (W/4)$ in turn. We splice the first upsampling result $Z_1 \in R^{256\times(H/8)\times(W/8)}$ and the second upsampling result $Z_2 \in R^{128\times(H/4)\times(W/4)}$ with the last layer and middle layer of the SRCB layer, respectively, to get the global and local information of the fusion feature map. At the skip connection between encoder and decoder, we propose the MASM, which employs three branches to enhance the selection and perception capabilities of low-level features, resulting in the feature tensor LO with reduced confusion information. To boost the segmentation impacts of edge pixels, the feature $X^l \in R^{C_i\times h\times w}$ obtained from the deepest SRCB is input into SIEM to gain the global clues extracted from the channel and spatial level to direct the decoder, and then employ the designed RGM to reencode the features from the encoder branch and the decoder branch to obtain feature $W_s \in R^{C_n\times H_n\times W_n}$ that enhances semantic information. In addition, we propose MFAM to compensate for the lack of detailed features in the upsampling process of the decoder and improve the segmentation effect of similar objects. We employ the fused features $P_1 \in R^{512\times(H/16)\times(W/16)}$, $P_2 \in R^{256\times(H/8)\times(W/8)}$, and $P_3 \in R^{128\times(H/4)\times(W/4)}$ of the three layers obtained by the decoder as the input of MFAM, and each layer obtains the feature map with the same scale and channel number through 3×3 convolution and two times upsampling. However, it is not enough to restore the spatial features. When the segmented objects share similar distribution patterns but different channels, it will easily lead to confusion in segmentation. Therefore, DCM is proposed to encode the correlation of channel dimensions and add it to the three-layer horizontal connection to get the outputs $O_t^s, O_p^s, O_n^s \in R^{128\times(H/2)\times(W/2)}$. Finally, these three features are fused, and then the final prediction mask is obtained by convolution and two times upsampling.

B. SIEM and RGM

Although the transformer has the advantage of obtaining global context information, it has defects in locating location information. One scheme is to encode the relative distance between tokens input in self-attention as in [33] and learn the pairwise relationship between tokens to solve the location problem. However, we explore a new scheme, which employs the advantages of CNN over transformer in location (translation invariance), designs SIEM to extract global clues from the channel and space level to guide the decoder, and then adopts the designed RGM to reencode the encoder branch and the decoder branch, so as to obtain the features of information enhancement and improve the segmentation ability of edge pixels. The structures of SIEM and RGM are shown in Figs. 2 and 3. In Algorithm 1, we describe in detail the process of obtaining an information enhancement feature map $W_s \in R^{C_n\times H_n\times W_n}$ by two algorithms, and we will introduce each part in detail below.

Algorithm 1 SIEM and RGM

Input: SRCB encoder output feature map X^l, Decoder branch feature map P_n, $n = f_1, f_2$;
Output: Output W_s of semantic information enhancement;

Part of SIEM:
$X' \leftarrow \rho(GN(DSC(X^l)))$;
$M_s \leftarrow Mean(X') + Max(X')$;
$M_c' \leftarrow \delta(MLP(AvgPool(X')) + MLP(MaxPool(X')))$;
$M_c \leftarrow \rho(GN(DSC(M_c')))$;
$X^{l+1} \leftarrow \delta(M_s \bigotimes M_c)$;
Part of RGM:
$u' \leftarrow Cat(P_n, interpolate(Conv_{1\times1}(X^{l+1})) \bigotimes P_n)$;
$W_s \leftarrow \zeta(BN(Conv_{3\times3}(u')))$;

The output feature map $X^l \in R^{C_i\times h\times w}$ from the last layer of the SRCB encoder is input to SIEM, and the channel is compressed by depthwise separable convolution (DSC) to obtain $X' \in R^{C_o\times h\times w}$, which aims to reduce the parameter quantity and further extract the deep semantic features. Then, the obtained features in the previous step are sent to two branches. One branch employs the average pooling function and the maximum pooling function to get the average and maximum value along the channel dimension to generate an effective feature descriptor, without processing the feature map and retaining the spatial information obtained in the previous step, the purpose of our design is to adopt the global information obtained by SRCB encoder to guide the decoder

$$X' = \rho\left(\mathrm{GN}\left(\mathrm{DSC}\left(X^l\right)\right)\right) \tag{1}$$

$$M_s = \mathrm{Mean}\left(X'\right) + \mathrm{Max}\left(X'\right) \tag{2}$$

where ρ represents the rectified linear unit (Relu) activation function and GN represents GroupNorm. In the other branch,

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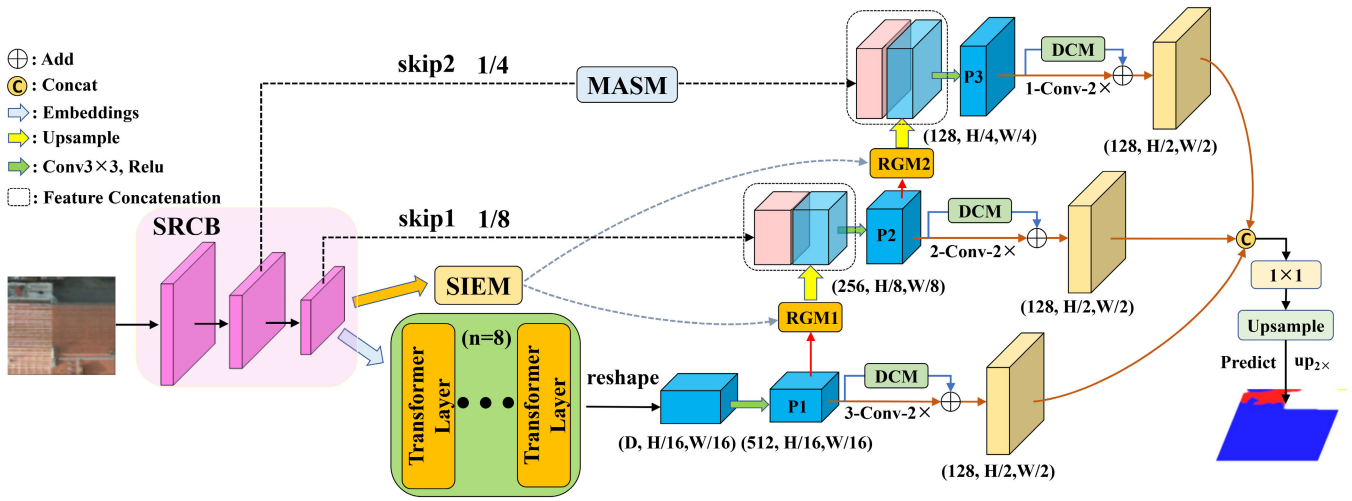


Fig. 1. Overall architecture of SRCBTFusion-Net comprises four main modules: 1) SIEM: extracting global cues to guide the decoder; 2) RGM: reducing the loss of detailed information, enhancing the discriminability of edge features and nonedge pixel features; 3) MASM: mitigating the confusion introduced by skip connections that fuse low- and high-level features; 4) MFAM: integrating multiscale features to enhance the extraction of semantic and contextual information, alleviating information loss and improving the discriminative ability for similar categories. The included DCM serves to compensate for the loss of detailed information during the upsampling process. SRCB represents SRCBs.

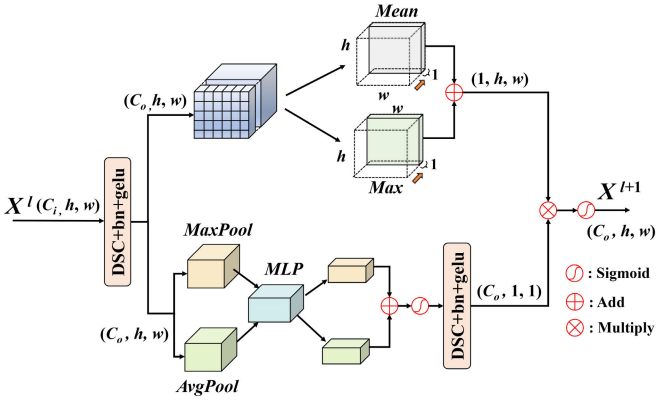


Fig. 2. SIEM.

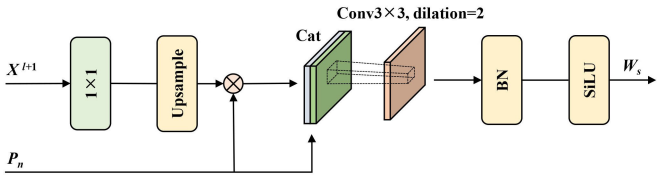


Fig. 3. RGM.

we employ global average pooling and global maximum pooling to compress global spatial information into channel descriptors, and then send them to multilayer perceptron (MLP) for information interaction. Some methods [6], [34] have shown that coding the association of channel dimensions can produce more discriminating features. Then, the attention weight matrix $M_c \in R^{C_o \times 1 \times 1}$ is obtained by DSC with batch normalization and Relu activation function

$$M_c' = \delta\left(\mathrm{MLP}\left(\mathrm{AvgPool}\left(X'\right)\right) + \mathrm{MLP}\left(\mathrm{MaxPool}\left(X'\right)\right)\right) \quad (3)$$

$$M_c = \rho\left(\mathrm{GN}\left(\mathrm{DSC}\left(M_c'\right)\right)\right) \quad (4)$$

where δ represents the sigmoid activation function. The calibrated attention feature map $X^{l+1} \in R^{C_O \times h \times w}$ is obtained by matrix multiplication between the attention weight M_c and output of the feature map M_s of another branch with spatial information

$$X^{l+1} = \delta(M_s \otimes M_c) \quad (5)$$

where $\otimes$ represents matrix multiplication. As shown in Fig. 3, The attention weight obtained by SIEM is input to the RGM. Because the number of channels and scale of the acquired feature map are different from those of the decoder, 1×1 convolution and bilinear interpolation are applied to unify the number of channels and the feature map scale, respectively, and then the obtained weights are weighted to the output feature map in the upsampling stage of the decoder to guide the decoder to get the features related to the edge area. To maintain the global context information gained by the Transformer and boost the detection capabilities of edge pixels, we employ stitching operation to fuse to get a feature map u'

$$u' = \mathrm{Cat}\left(P_n, \mathrm{interpolate}\left(\mathrm{Conv}_{1\times1}\left(X^{l+1}\right)\right) \otimes P_n\right) \quad (6)$$

where Cat represents the splicing operation, and P_n represents the characteristic diagram at the position $n = \{f_1, f_2\}$ in the upsampling stage of the decoder, interpolate represents upsampling, and $\mathrm{Conv}_{1\times1}$ represents convolution with convolution kernel 1×1. To gain a wider range of information, we adopt the dilated convolution $\mathrm{Conv}_{3\times3}$ (convolution with an expansion rate of 2 and convolution kernel of 3×3) to reencode the fusion features and then normalize them to get the feature map W_s with enhanced semantic information

$$W_s = \zeta\left(\mathrm{BN}\left(\mathrm{Conv}_{3\times3}\left(u'\right)\right)\right) \quad (7)$$

where ζ represents the sigmoid linear unit (SiLU) activation function, which is smoother than the Relu activation function, and there are cases where the value is negative, so more information can be retained.

C. Multipath Atrous Self-Attention Module

Inspired by [35] and [22], to address the potential confusion generated by the skip connections between fused low- and high-level features, we propose the MASM. This module aims to enhance the specificity of low-level feature contributions by

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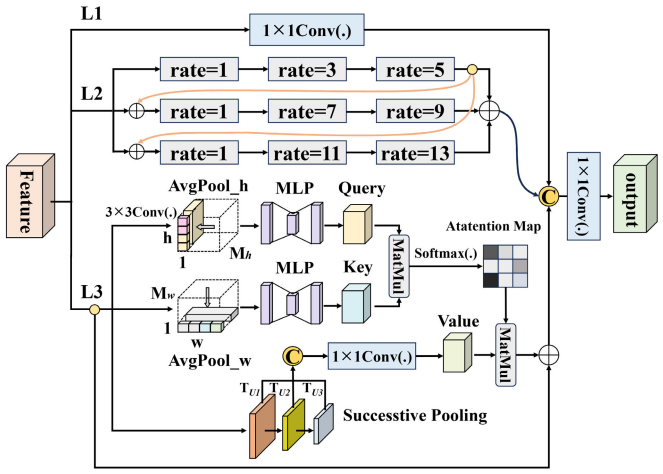


Fig. 4. MASM.

Algorithm 2 MASM

Input: Encoder output low-level feature $BO_d \in R^{C_m \times H_m \times W_m}$;
Output: Reduce the confusing information fusion feature $LO \in R^{C_m \times H_m \times W_m}$;

Branch1:
$CV_a \leftarrow \rho(BN(Conv_{1\times1}(BO_d)))$;
Branch2:
The outputs F_1, F_2, F_3 of the three branches are obtained in turn by Equation (9), and PU_b is obtained after fusing them:
$PU_b \leftarrow F_1 \oplus F_2 \oplus F_3$;
Branch3:
Adopting the Equation (11),(12),(13),(14),(15),(16),(17) complete the initialization of Query, Key and Value, and attention map AQ_c is obtained by Equation (18) and (19);
The output of the fusion of three branch:
$LO \leftarrow Conv_{1\times1}(Cat(CV_a, PU_b, AQ_c))$

effectively selecting and weighting them, ensuring improved capture of local details. To handle the introduction of weak semantic information, we design a self-attention mechanism that, while preserving low-level features, enables more precise selection and integration, thereby reducing the introduction of confusion. In addition, a stacked dilated convolution block is designed to enhance the network's perception of broader contextual information. By dynamically adjusting the attention given to features at different levels, the model can flexibly capture information at various semantic levels, selectively reinforcing the learning of low-level semantics. See Algorithm 2 for the algorithm implementation of MASM.

The network structure of MASM is shown in Fig. 4, comprising three branches: L1, L2, and L3. Initially, similar to ASPP, we introduce a 1×1 convolution to reduce dimensionality as one branch, followed by an activation layer to introduce nonlinear activation. Fusion with other branches enhances the performance of segmentation tasks

$$CV_a = \rho(\mathrm{BN}(\mathrm{Conv}_{1\times1}(\mathrm{BO}_d))). \tag{8}$$

L2 represents the second branch, which employs a three-layer stacked dilated convolution block (convolution kernel of 3×3 with different dilation rates). The output of the nth layer is represented as follows:

$$F_n = \begin{cases} S_n(\mathrm{BO}_d, \theta_n), & n = 1 \\ S_n(\mathrm{BO}_d \oplus F_1, \theta_n), & n = 2, 3 \end{cases} \tag{9}$$

where $\oplus$ represents that addition of the corresponding element in the tensor. S_n represents the output of each branch and θ_n is a learnable parameter. $\mathrm{BO}_d \in R^{(C_m/N)\times H_m \times W_m}$ represents the output of the encoder, where we compress the input channels by a factor of $N = 16$ and restore them in the final convolutional block. Subsequently, the outputs from the three branches are fused to obtain feature PU_b

$$\mathrm{PU}_b = F_1 \oplus F_2 \oplus F_3. \tag{10}$$

The L3 branch, inspired by [35] and self-attention [36], incorporates improvements upon self-attention. This modification enables the network to dynamically adjust the importance of different positions in the feature maps, contributing to an enhanced grasp of both local details and global semantics. Similar to the input of the L2, the input of L3 is also assumed to be $\mathrm{BO}_d \in R^{C_m \times H_m \times W_m}$, differing in the initialization approach for Query (Q), Key (K), and Value (V) from [36]. First, the input BO_d is passed through a convolution with a kernel size of 3 to obtain the tensor N_d. Subsequently, adaptive average pooling is utilized to obtain statistics for Q in the vertical spatial direction. The obtained K is similarly acquired through adaptive average pooling but in the horizontal spatial direction. The specific calculation formula is as follows:

$$M_h = \frac{1}{W_m} \sum_{j=0}^{W_m-1} \hat{N}_d(i, j) \tag{11}$$

$$M_w = \frac{1}{H_m} \sum_{i=0}^{H_m-1} \mathrm{BO}_d(i, j) \tag{12}$$

where i and j are indexes in vertical direction and horizontal direction. Here, $0 \le i < H_m, 0 \le j < W_m, \hat{N}_d = \mathrm{Conv}_{3\times3}(N_d)$. Therefore, $M_h \in R^{C_m \times H_m \times 1}$ and $M_w \in R^{C_m \times 1 \times W_m}$ are obtained by (11) and (12), to enhance information exchange between channels, they are separately fed into an MLP layer with two 1×1 convolutions to obtain $Q \in R^{C_m \times H_m \times 1}$ and $K \in R^{C_m \times 1 \times W_m}$

$$Q = \mathrm{MLP}(M_h) \tag{13}$$

$$K = \mathrm{MLP}(M_w). \tag{14}$$

The initialization of V involves executing three consecutive pooling operations to gather features at different scales. This transforms input feature $\mathrm{BO}_d \in R^{C_m \times H_m \times W_m}$ into feature $T_{U_i} \in R^{C_m \times PH_i \times \mathrm{PW}_i}, i = 1, 2, 3$. Specifically, BO_d undergoes three consecutive downsampling operations and 1×1 convolution operations (with normalization and ReLU activation) to produce $T_{U_1} \in R^{C_m \times (H_m/2) \times (W_m/2)}$, $T_{U_2} \in R^{C_m \times (H_m/6) \times (W_m/6)}$, and $T_{U_3} \in R^{C_m \times (H_m/8) \times (W_m/8)}$, respectively. Afterward, T_{U_1}, T_{U_2}, and T_{U_3} are upsampled to the same scale. After concatenation and a 1×1 convolution operation, an enhanced feature matrix V is obtained to strengthen contextual connections

$$T_{U_i} = \frac{1}{H_m \times W_m} \sum_{x=1}^{H_m} \sum_{y=1}^{W_m} P(x, y) \tag{15}$$

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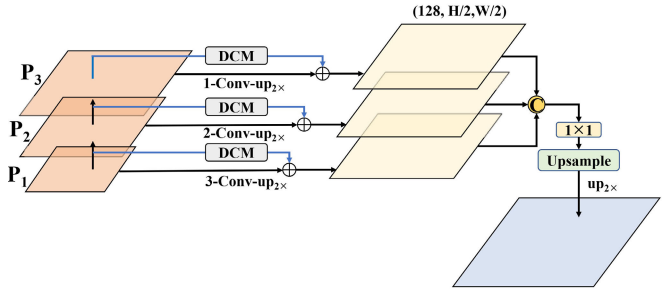


Fig. 5. MFAM.

$$PV_i = \text{interpolate}\left(\rho\left(\text{BN}\left(\text{Conv}_{1\times1}\left(T_{U_i}\right)\right)\right)\right) \tag{16}$$

$$V = \text{Conv}_{1\times1}(\text{Cat}(PV_i)). \tag{17}$$

The initialization of Q, K, and V is completed with the aforementioned operations. Subsequently, the procedure follows [36], where $d_{\text{head}} = 128$ in this article. To enhance the stability of gradient backpropagation, we introduce a residual structure. The calculation is as follows:

$$\text{AT} = \text{Softmax}\left(\frac{QK}{\sqrt{d_{\text{head}}}}\right)V \tag{18}$$

$$\text{AQ}_c = \text{AT} \oplus \text{BO}_d. \tag{19}$$

Finally, the features obtained from the three branches are fused

$$\text{LO} = \text{Conv}_{1\times1}(\text{Cat}(CV_a, \text{PU}_b, \text{AQ}_c)). \tag{20}$$

D. Multiscale Feature Aggregation Module

On the one hand, to further narrows the gap between local and global information because of fusing the encoder and decoder, on the other hand, it alleviates the loss of detailed information in the continuous upsampling process of the decoder. We propose a new multiscale feature fusion structure called MFAM. Different from the prediction based on the deepest layer in [11] and [37], we regard the independent feature mapping of each layer of the decoder as equally important. After step-by-step upsampling and a detailed information supplement, we obtain outputs rich in global context information and fine-grained spatial features. Then, the effective fusion strategy is adopted to predict. Fig. 5 shows the network structure of MFAM. See Algorithm 3 for the algorithm implementation of MFAM.

Algorithm 3 MFAM

Input: Multiscale feature map L^s, $s = \alpha_1, \alpha_2, \alpha_3$;
Output: Output Multiscale fusion output V^m;

$O_t^s \leftarrow 3 \times up_{2\times}(TConv(L^{\alpha_1})) + DCM(L^{\alpha_1})$;
$O_p^s \leftarrow 2 \times up_{2\times}(TConv(L^{\alpha_2})) + DCM(L^{\alpha_2})$;
$O_n^s \leftarrow 1 \times up_{2\times}(TConv(L^{\alpha_3})) + DCM(L^{\alpha_3})$;
Fusion output:
$V^m \leftarrow up_{2\times}(Conv_{1\times1}(Cat(O_t^s, O_p^s, O_n^s)))$;

First, the extracted feature L^s of the decoder is delivered to MFAM. The output of the sth layer is specified as follows:

$$L^s = \left\{l^{a_1}, l^{a_2}, l^{a_3}\right\}. \tag{21}$$

Proceeding from the bottom layer of the decoder, we perform three step-by-step upsampling operations and make a smooth transition by convolution after each upsampling. Finally, we obtain a feature map with a number of channels is 128 and the size is 1/4 of the original image resolution. The purpose of upsampling step by step is to recover spatial detail information, and the stacking times of convolution blocks decrease step by step as other layers go up. Three-layer independent feature representation is as follows:

$$O_t^s = 3 \times \text{up}_{2\times}\left(\text{TConv}\left(L^{a_1}\right)\right) + \text{DCM}\left(L^{a_1}\right) \tag{22}$$

$$O_p^s = 2 \times \text{up}_{2\times}\left(\text{TConv}\left(L^{a_2}\right)\right) + \text{DCM}\left(L^{a_2}\right) \tag{23}$$

$$O_n^s = 1 \times \text{up}_{2\times}\left(\text{TConv}\left(L^{a_3}\right)\right) + \text{DCM}\left(L^{a_3}\right) \tag{24}$$

where O_t^s represents the feature map of the sth layer at a spatial position $t = \{1, 2, 3\}$, O_p^s represents the feature map of the sth layer at a spatial position $t = \{1, 2\}$, and O_n^s represents the feature map of the sth layer at a spatial position $t = \{1\}$. $\text{up}_{2\times}$ represents two times bilinear upsampling. TConv represents the combination structure of convolution (padding and stride size of 1 and convolution kernel of 3×3) and BatchNorm and SiLU activation function.

In addition, to compensate for the lack of detailed features in the upsampling process and further improve the segmentation ability of similar objects, DCM is proposed as shown in Fig. 6, and the algorithm implementation is shown in Algorithm 4. Here, we adopt global average pooling to compress a resolution $h \times w$ feature map with c channels into a resolution 1×1 feature map Z_c with c channels

$$Z_c = \frac{1}{H \times W}\sum_{i=1}^{H}\sum_{j=1}^{W} P(i, j). \tag{25}$$

To overcome the issue of confusing segmentation when the segmented objects share similar distribution patterns but different channels. We send the previous step obtained feature map into a two-layer 1×1 neural network (the stride of the first layer is 1, and the stride of the second layer is 2) to learn the weight of each channel, so as to improve the display modeling ability of the channel dimension relationship [38]. The equation of the weight matrix $T \in R^{C_i \times 1 \times 1}$ is as follows:

$$T_{\text{down}} = \zeta\left(\text{BN}\left(W_{1(\text{GN}(Z_c))}\right)\right) \tag{26}$$

$$T = W_2 T_{\text{down}} \tag{27}$$

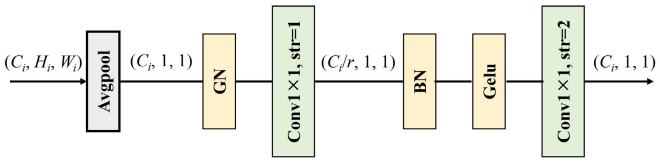


Fig. 6. DCM.

Algorithm 4 DCM

Input: Arbitrary four-dimensional image P;
Output: Characteristic diagram T of channel correlation enhancement;

$Z_c \leftarrow F(P) = \frac{1}{H \times W}\sum_{i=1}^{H}\sum_{j=1}^{W} P(i, j)$;
$T_{down} \leftarrow \zeta(BN(W_1(GN(Z_c))))$;
$T \leftarrow W_2 T_{down}$;

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where $W_1 \in R^{(C_i/r)\times C_i}$ represents decreasing the channel dimension. $W_2 \in R^{C_i\times(C_i/r)}$ represents the reduction channel dimension. $r = \{2, 4, 8\}$ represents the channel reduction multiple, which increases from the lower level to the upper level in turn. We try to add the DCM module to three transverse branches of MFAM by matrix multiplication and summation and find that the summation effect is the best. Finally, the three branches with the same scale and channel numbers are fused by stitching strategy, and then the prediction label V^m of each type of pixel is generated by 1×1 convolution and two times upsampling to the original image size

$$V^m = \mathrm{up}_{2\times}\left(\mathrm{Conv}_{1\times1}\left(\mathrm{Cat}\left(O_t^s, O_p^s, O_n^s\right)\right)\right). \tag{28}$$

IV. Experiment Results and Analysis

A. Datasets

In our experiment, all the results of the Vaihingen dataset and Potsdam dataset (https://www.isprs.org/education/benchmarks/UrbanSemLab/default.aspx) are tested only by employing the true orthophotos (TOP) image block.

1) Potsdam Dataset: Potsdam dataset contains 38 very fine spatial resolution TOP image blocks (ground sampling distance (GSD) is 5 cm) with a size of 6000×6000 pixels. Four multispectral bands (red, green, blue, and near-infrared), digital surface models (DSMs), and normalized DSMs (NDSMs) are provided in the dataset. Only three bands (red, green, and blue) are adopted, DSMs and NDSMs data are not included in the experiment. Same as [39] and [11], we adopt ID: 2_13, 2_14, 3_13, 3_14, 4_13, 4_14, _15, 5_ 13, 5_14, 5_15, 6_14 for testing and the remaining 24 images are utilized for training. The original image block is cropped to 256×256.

2) Vaihingen Dataset: The Vaihingen dataset contains 33 TOP images collected by advanced airborne sensors. The GSD is about 9 cm. Each TOP image has three channels: infrared, red, and green, and has the same category information as the Potsdam dataset. After [40] and [11], we select ID:1, 3, 5, 7, 13, 17, 21, 23, 26, 32, 37 for training and ID: 11, 15, 28, 30, 34 for testing and crop them to 256×256.

B. Experimental Setup

1) Experimental Parameter Settings: We adopt a series of data expansion strategies in the training dataset. For example, random rotation (90°, 180°, and 270°), flip along X- and Y-axis, gamma transform, mean filtering, bilateral filtering, Gaussian filtering, and adding noise. The ratio of all policies is set to 0.25. Although increasing the image size can improve the segmentation performance, in order to meet the requirements of the hardware environment, the image size is adjusted to 256×256, the batch size is set to 16, the learning rate is adjusted by cosine annealing strategy, the initial learning rate is set to $2e-4$, and the minimum learning rate is set to $1e-4$. All models employ the AdamW optimization strategy. To alleviate the over-fitting phenomenon and save unnecessary training time, all experiments adopt the early stop method (the maximum period is 60, and the early stop period is 10), and the weight attenuation is set to $5e-4$. We employ the pretraining weights provided by Chen et al. [10] to initialize the backbone.

TABLE I
Ablation Experiment of Loss Function on Potsdam Dataset

Model	Loss Function			Evaluation index		
	CE	DTL	MCDL	MIoU (%)	Ave.F1 (%)	Time (m)
SRCBTFusion-Net	√			78.14	87.59	4.2
	√	√		**78.62**	**87.98**	4.5
	√	√	√	78.34	87.73	7.2

a) Note: In the later experiments, there is no special explanation that the number of stacked layers of the transformer is eight. In the following tables, bold represents the highest accuracy. In the following figures, GT represents ground truth.

2) Evaluation Metrics: To evaluate the accuracy of SRCBTFusion-Net, the mean intersection over union (MIoU) and average $F1$ (Ave.$F1$) score are employed for evaluation. and these scores are calculated according to the cumulative confusion matrix. In addition, evaluating the scale of the network includes the number of floating point operations (Flops) to assess the complexity, the number of frames per second (FPS) to assess the speed, the unit minutes of model training time (m) and the number of model parameters (M).

3) Loss Function: To avoid the loss function dominated by nonboundary pixels when the boundary pixels do not match the actual boundary, the loss function of DualTaskLoss [41] (DTL) is introduced. In addition, we also introduce CrossEntropyLoss (CE) same as [11] to evaluate the accuracy of semantic segmentation. To ensure the rationality of the selected loss function. Table I shows the ablation experiments on the above two kinds of losses and the loss of MultiClassDiceLoss (MCDL), and finds that the combined loss of CE and DTL has the highest accuracy in MIoU and Ave.$F1$, and the training time is less than 1 min longer than that of CE only. Of course, considering the proportion between the two loss functions, we conduct experiments on the weight coefficients of the two functions and find that 1:1 is the best. Therefore, the common loss L is expressed as follows:

$$L = L_{\mathrm{CE}} + L_{\mathrm{DTL}}. \tag{29}$$

4) Hardware Platform: All experiments are implemented on a workstation with 64-GB memory Intel i9-11900K at 3.50-GHz CPU and 24-GB NVIDIA GeForce RTX 3090 GPU.

C. Research on the Rationality of Structural Design

We analyze and discuss the rationality of the proposed modules. ***Note:*** The combination of SIEM and RGM is abbreviated as SR. We will get the network structure in Fig. 1 from the original backbone structure in Fig. 7 in the following discussion. The backbone network without any modules is replaced by SRTFNet$_{x_2}$, and the backbone network with SR and MFAM is replaced by SRTFNet$_{x_3}$.

1) Effectiveness of SR Module: Based on the model proposed by [10], we explore the combination of adding four SR modules to enhance semantic information and minimize the absence of detailed information. We do not add the designed MFAM module to avoid interfering with the experiment and adapt the output of the top layer to get the final prediction result. We perform six groups of tests, respectively, in Table II, and the greatest MIoU and Ave.$F1$ are 77.49% and 87.19%

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TABLE II
PERFORMANCE ANALYSIS OF DIFFERENT COMBINATIONS OF SR MODULES ON POTSDAM DATASET

Model	IoU (%)						Evaluation index	
	Impervious Surface	Building	Low Vegetation	Tree	Car	Clutter	MIoU (%)	Ave.F1 (%)
Backbone	79.21	86.12	69.87	70.44	78.66	30.48	76.86	86.76
Backbone + SR1 + SR2	**79.39**	86.30	69.70	71.49	79.96	30.63	77.36	87.10
Backbone + SR1 + SR3	79.12	86.13	69.91	71.55	78.98	29.00	77.14	86.97
Backbone + SR2 + SR3	**79.39**	**86.54**	69.92	70.86	80.21	**31.53**	77.38	87.11
Backbone + SR2 + SR3 + SR4	79.28	86.21	**70.08**	71.01	79.54	30.43	77.22	87.02
Backbone + SR1 + SR2 + SR3	79.24	86.33	69.68	71.59	80.26	29.70	77.42	87.14
Backbone + SR1 + SR2 + SR3 + SR4	79.24	86.24	69.72	**71.92**	**80.34**	30.30	**77.49**	**87.19**

TABLE III
EXPLORING THE SEGMENTATION PERFORMANCE OF DIFFERENT COMBINATIONS OF MFAM ON THE POTSDAM DATASET

Model	IoU (%)						Evaluation index	
	Impervious Surface	Building	Low Vegetation	Tree	Car	Clutter	MIoU (%)	Ave.F1 (%)
SRTFNet$_{x_1}$ + P3	79.61	86.55	69.75	72.24	79.74	31.21	77.58	87.21
SRTFNet$_{x_1}$ + P2 + P3 (cat)	79.73	86.84	70.28	**72.37**	80.04	31.36	77.85	87.36
SRTFNet$_{x_1}$ + P1 + P2 + P3 (cat)	79.60	**87.62**	**70.76**	71.91	**80.69**	31.45	**78.12**	**87.58**
SRTFNet$_{x_1}$ + P1 + P2 + P3 (add)	**79.93**	87.02	70.21	71.86	80.66	**32.04**	77.93	87.46
SRTFNet$_{x_1}$ + P3 + P4 (cat)	79.58	86.86	69.85	72.30	80.32	30.78	77.78	87.37
SRTFNet$_{x_1}$ + P2 + P3 + P4 (cat)	79.39	86.88	70.14	71.49	80.33	31.76	77.65	87.28
SRTFNet$_{x_1}$ + P1 + P2 + P3+ P4 (cat)	79.33	87.06	70.74	72.21	80.56	31.34	77.98	87.50

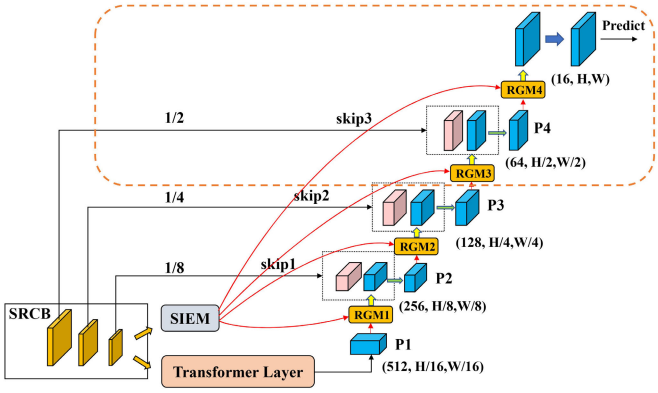


Fig. 7. Original backbone network structure.

in combination with four SRs, which are 0.63% and 0.43% higher than the backbone network. In addition, from the data of rows 3, 4, and 5, it can be analyzed that the combination of SR2 and SR3 is particularly prominent in improving the segmentation performance. Based on this combination, increasing SR1 performance has a slight improvement, while increasing SR4 has a poor effect. This is because the combination of SR2 and SR3 enhances the middle- and deep-level semantic information and further refines the features. However, the fusion features of SR4 act on SRCB's local semantic information and the global semantic information processed in the decoding stage. Because of the significant difference in spatial and location information, it is often impossible to effectively alleviate the difference. So we try to combine [SR1/SR2/SR3/SR4], and find that the segmentation effect is improved, which shows that the addition of SR1 eases the gap between the output of the encoder and SRCB's local semantics, and combines with SR4 to compensate for the loss of detailed information.

Second, we also find that the addition of the SG module significantly improves the segmentation accuracy of "Tree" (increased by 1.48%) and "Car" (increased by 1.68%) contrasted with the backbone, which shows that our proposed model has good segmentation effect for small target objects.

2) Study on Structural Rationality of MFAM: Different from [10], we assume that fusing the features of each decoder layer is beneficial to fully capture the detailed information and further enhance the segmentation capacity for targets with varied scales. The backbone network is named SRTFNet$_{x_1}$ with [SG1/SG2/SG3/SG4] combination, and the top layer is adopted for prediction. In order to explore which combination method can achieve the best segmentation accuracy, we compare seven methods in Table III. Among them, the best MIoU (78.12%) and Ave.$F1$ (87.58%) are achieved by merging the outputs of the P1, P2, and P3 layers, while the effect is slightly worse by adding the outputs. This is because the latter will cause feature overlap. In addition, we try to combine the top layers of the decoder. In Table III, the third row compared to the last row, it is found that adding the P4 layer will greatly reduce the segmentation accuracy of "Impervious Surface," "Building," and "Tree." This is because of the tremendous semantic difference after the introduction of skip3, and the structural design of this model can't effectively filter the rough information after splicing, so we discard the structure in the dotted box in Fig. 7, which minimizes the number of model params and improves the precision of segmentation.

3) Study the Number of Layers of the Transformer: To better extract global information from the output of SRCB by transformer, the number of cycles becomes the key factor affecting the effect. In Table IV, we test the effects of zero, five, eight, and 12 stacks, respectively, and the best performance is eight cycles. With the growth of the number of cycles, the training time becomes longer while the number of parameters increases, but the segmentation effect becomes worse. This is because the deeper the network, the more difficult it is to train and optimize, and the deeper the network, the more degraded it will be. Therefore, the transformer cycle number is set to eight.

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TABLE IV
EXPLORING THE INFLUENCE OF TRANSFORMER STACK LAYERS ON POTSDAM DATASET

Layer	Params (M)	IoU (%)						Evaluation index		
		Impervious Surface	Building	Low Vegetation	Tree	Car	Clutter	MIoU (%)	Ave.F1 (%)	Time (m)
0	29.59	77.80	84.42	69.52	69.76	80.14	28.80	76.33	86.45	3.5
5	53.12	80.35	87.07	70.63	72.46	81.64	31.36	78.43	87.78	4.5
8	86.30	**80.50**	**87.23**	**71.17**	**72.52**	**81.66**	**33.15**	**78.62**	**87.98**	4.5
12	114.65	80.04	**87.23**	70.36	71.99	80.66	29.51	78.06	87.54	4.9

TABLE V
CONDUCT ABLATION EXPERIMENTS OF SIEM, RGM, AND DCM ON THE POTSDAM DATASET. w/o MEANS WITHOUT

Model	IoU (%)						Evaluation index	
	Impervious Surface	Building	Low Vegetation	Tree	Car	Clutter	MIoU (%)	Ave.F1 (%)
SRTFNet$_{x_2}$ + SIEM	79.33	86.10	69.56	71.19	79.20	30.23	77.08	86.91
SRTFNet$_{x_2}$ + RGM	79.35	86.21	69.65	71.26	79.43	30.36	77.18	86.97
SRTFNet$_{x_2}$ + SR + MFAM (w/o DCM)	79.49	87.23	70.36	71.74	80.24	30.79	77.81	87.33

D. Ablation Experiment of Structure

In this section, we adopt the structure in Fig. 1 as SRCBTFusion-Net to evaluate the efficacy of the vital components. ***Note:*** The combination of SIEM and RGM is abbreviated as SR.

1) Effect Analysis of SR Module: First, let's discuss the contributions of the SIM and RGM modules in the SR module. As shown in Table V, the MIoU of the network with SIM and RGM improved by 0.28% and 0.38%, respectively, compared with the backbone SRTFNet$_{x_2}$. The incorporation of both modules shows significant enhancements for categories such as "Car" and "Tree," particularly for small-scale and edge-blurred instances. This observation indicates a strong capability to capture detailed information. According to the results in Table VI, when adding the SR module in SRCBTFusion-Net, the model parameters increased by 6.03M and the evaluation index increased by 0.56% (MIoU) and 0.38% (Ave.$F1$). Specifically, the most increased category is "Car" (increased by 1.41%), followed by "Tree" (increased by 1.13%). Combined with the analysis of the visualization results in Fig. 8, in the first row, the boundary between "Low Vegetation," "Car," and "Impervious Surface" is more apparent after the SR module is added, especially the boundary continuity of "Low Vegetation" class is enhanced. From the third row, we can see that SRCBTFusion-Net has an incomplete segmentation of the "Building" class, which is due to the lack of an effective edge detection strategy. After adding SR in the second and third rows, the boundary of "Building" is smoother and the segmentation accuracy is improved. At the same time, it also proves that our proposed edge detection scheme is effective.

2) Effect Analysis of the Combination of SR and MFAM Modules: Because the advantages of MFAM cannot be known after adopting it alone, we combine SR and MFAM. From Table VI, the evaluation index increased by 0.76% (MIoU) and 0.48% (Ave.$F1$) compared with adding SR alone, and the IoU in all six categories is improved. This proves that the designed MFAM has advantages in integrating multiscale features and reducing detailed information. According to the visualization results in Fig. 9. In the first row, the two types of buildings are distinguished together after adding SR alone, ignoring "Low Vegetation," which is caused by the small overall area of the

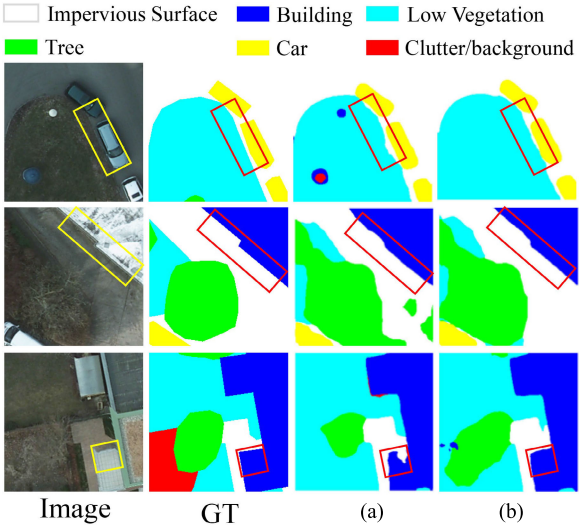


Fig. 8. Comparison of segmentation effect of backbone network before and after employing SR. (a) SRTFNet$_{x_2}$. (b) SRTFNet$_{x_2}$+SR.

class "Low Vegetation" in the figure. However, after adding MFAM, "Low Vegetation" and "building" are distinguished. Similarly, in the second and third rows, the "LowVegetation" class is separated by "Tree," which proves that the proposed MFAM module is beneficial for distinguishing similar objects.

Furthermore, we also discuss the contribution of DCM in MFAM. As seen in Tables VI and V, the MIoU and Ave.$F1$ for the tests without DCM exhibit a decrease of 0.31% and 0.25%, respectively. Particularly, the categories "Low Vegetation" (decreased by 0.40%) and "Car" (decreased by 0.45%) show the most significant declines. The former category often overlaps with the "Tree" category, leading to confusion, while the latter category, due to its small size and close proximity, is challenging to segment. The inclusion of DCM alleviates these issues, enhancing discriminability both between and within categories.

3) Performance Comparison and Analysis of FPN and MFAM Modules: FPN [42] has an excellent performance in the task of target detection and has the advantage of distinguishing complex targets. To demonstrate the advantages of our proposed multiscale fusion strategy, we keep the experimental parameters the same and only replace MFAM

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TABLE VI
ABLATION EXPERIMENT OF THE DESIGNED MODULES ON THE POTSDAM DATASET

Backbone	Params (M)	Modules					IoU (%)						Evaluation index	
		SR	MFAM	MASM	FPN	ASPP	Impervious Surface	Building	Low Vegetation	Tree	Car	Clutter	MIoU (%)	Ave.F1 (%)
SRTFNet$_{x_2}$	76.62						79.23	86.02	69.87	70.36	78.55	30.25	76.80	86.72
	82.65	√					79.39	86.30	69.70	71.49	79.96	30.78	77.36	87.10
	84.94	√			√		79.42	87.21	70.55	71.85	79.50	29.43	77.71	87.33
	84.94	√	√				79.60	87.62	70.76	71.95	80.69	31.07	78.12	87.58
	86.30	√	√	√			80.50	87.23	71.17	72.52	81.66	33.15	78.62	87.98
	84.91	√	√			√	79.76	87.38	70.91	72.51	80.75	30.63	78.26	87.68

TABLE VII
CONDUCT ABLATION EXPERIMENTS ON THE INTERNAL STRUCTURE OF MASM ON THE POTSDAM DATASET

Model	IoU (%)						Evaluation index	
	Impervious Surface	Building	Low Vegetation	Tree	Car	Clutter	MIoU (%)	Ave.F1 (%)
SRTFNet$_{x_3}$ + L1 + L2	79.68	86.97	70.92	72.39	81.45	30.62	78.28	87.66
SRTFNet$_{x_3}$ + L1 + L3	80.21	**87.25**	70.56	**72.59**	81.24	30.85	78.37	87.76
SRTFNet$_{x_3}$ + L1 + L2 + L3	**80.50**	87.23	**71.17**	72.52	**81.66**	**33.15**	**78.62**	**87.98**

TABLE VIII
DISCUSSING THE PLACEMENT OF MASM ON SKIP CONNECTIONS ON THE POTSDAM DATASET

Model	IoU (%)						Evaluation index	
	Impervious Surface	Building	Low Vegetation	Tree	Car	Clutter	MIoU (%)	Ave.F1 (%)
SRCBTFusion-Net with MASM$_1$	79.72	86.78	71.00	72.33	80.62	30.80	78.09	87.58
SRCBTFusion-Net with MASM$_2$	**80.50**	87.23	**71.17**	72.52	**81.66**	**33.15**	**78.62**	**87.98**
SRCBTFusion-Net with MASM$_1$ and MASM$_2$	79.84	**87.58**	70.24	**72.63**	81.40	29.65	78.34	87.72

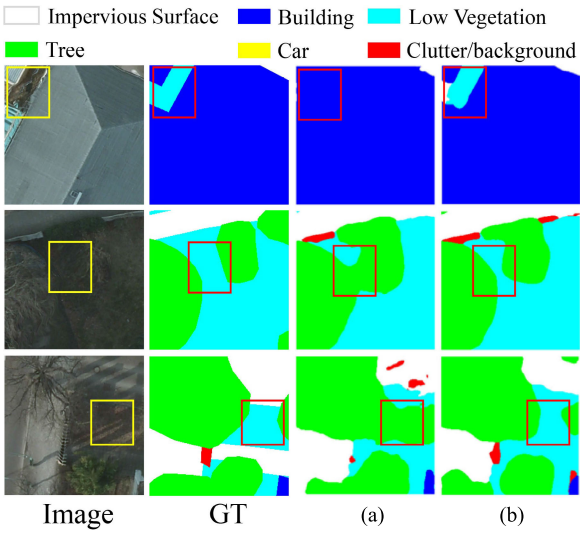


Fig. 9. Comparison of segmentation effect of backbone network before and after employing SR and MFAM. (a) SRTFNet$_{x_2}$+SR. (b) SRTFNet$_{x_2}$+SR+MFAM.

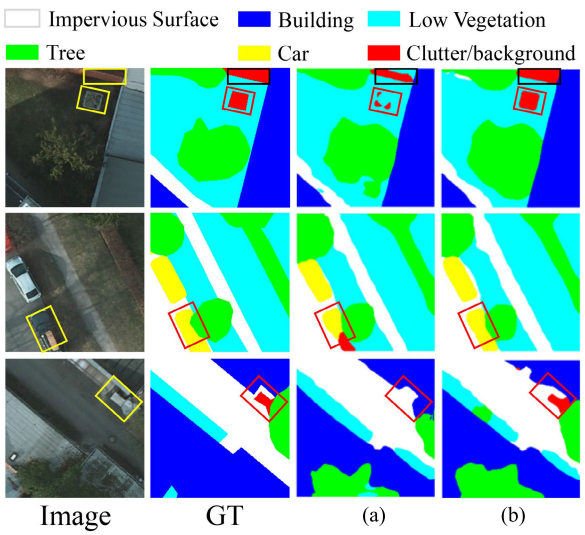


Fig. 10. Comparison of segmentation results between two multiscale feature fusion modules. (a) SRTFNet$_{x_2}$+SR+FPN. (b) SRTFNet$_{x_2}$+SR+MFAM.

with the FPN module. In Table VI, The SRCBTFusion-Net with MFAM and SRCBTFusion-Net with FPN have the same parameters of 2.29M, but the former is 0.41% and 0.25% higher than the latter in MIoU and Ave.$F1$. The visualization result is presented in Fig. 10, in the first and third rows, SRCBTFusion-Net clearly distinguishes the "Background" of the small area completely. Besides, in the second row, although the small-sized "Car" is partially covered by "Tree," SRCBTFusion-Net can still accurately segment the semantic area. This shows that the structural design of MFAM is superior to FPN, and it is advantageous to boost the segmentation precision of small-scale targets.

4) Effect Analysis of Adding MASM to the Backbone SRTFNet$_{x_3}$: The experimental results in Table VII serve as evidence demonstrating the rationality of the structural design of MASM. Given that the primary role of the top 1×1 convolutional operation is to introduce nonlinearity and enhance the capability to handle complex scenes, similar to the effects of [22], it requires combination with other model components. Therefore, our focus is on exploring the contributions of its combination with the designed branches of L2 and L3. The results indicate a significant improvement in segmentation accuracy with the inclusion of L2 and L3 branches. The L3 branch, utilizing a self-attention structure, effectively captures

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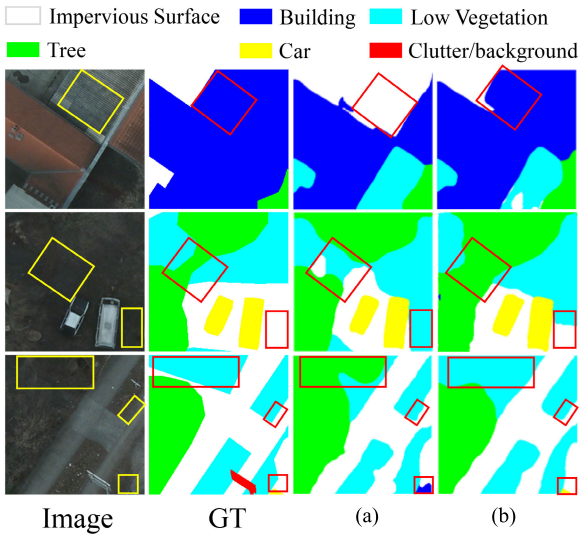


Fig. 11. Comparison of segmentation results before and after adding MASM. (a) SRTFNet$_{x_3}$. (b) SRTFNet$_{x_3}$+MASM.

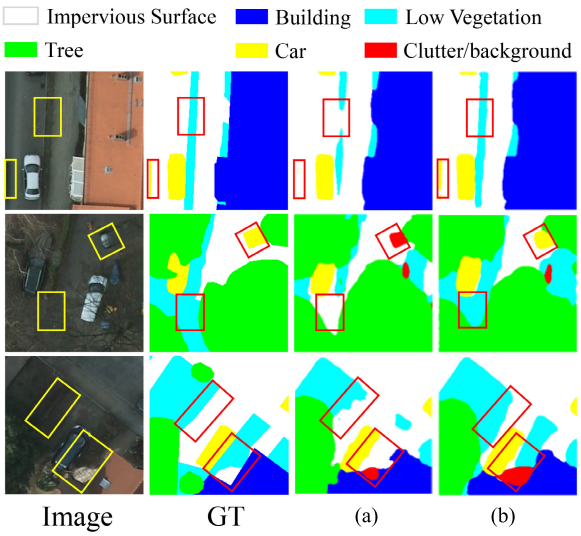


Fig. 12. Segmentation results comparison before and after replacing MASM with ASPP. (a) SRTFNet$_{x_3}$+ASPP. (b) SRTFNet$_{x_3}$+MASM.

local details and global semantics. Consequently, it proves particularly effective for categories like "Building" and "Tree" that occupy a large extent in the image. On the other hand, the L2 branch employs stacked convolutional blocks with different receptive fields, enhancing the network's perception of broader contextual information. When the three branches are fused, MIoU and Ave.$F1$ improve by 0.50% and 0.40%, respectively, with only a marginal increase in parameters (1.36M), demonstrating the superiority of the design structure. As shown in Table VIII, we discuss the positions where MASM is added and observe that its application to skip2 yields the best segmentation results. This is attributed to the fact that lower layers possess highly detailed feature maps with greater resolution, making them particularly helpful for accurate localization of the target positions. The visual results are shown in Fig. 11. In the first row, after adding the MASM module, the segmentation for the "Building" category becomes more complete. In addition, in the second and third rows, there is an enhancement in the discriminability between the categories "Tree" and "Low Vegetation," as well as between "Impervious Surface" and "Low Vegetation."

5) Performance Comparison Analysis of Adding MASM and ASPP Modules on the Backbone SRTFNet$_{x_3}$: ASPP [22] has the advantage of acquiring a broad receptive field, particularly excelling in compensating for detailed information and detecting small targets. However, the complexity of scenes limits the application of ASPP in the segmentation domain, and in this study, there is a need to handle low-level features for fusion with high-level features. Hence, we design the MASM. To understand the advantages of the improved MASM over ASPP, the results of replacing MASM with ASPP in Table VI. The parameter increases by 0.61M, while MIoU and Ave.$F1$ decrease by 0.36% and 0.30%, respectively. Visual results in Fig. 12 show that, in the first and second rows, the network with added MASM has an advantage in capturing small areas, and in the third row, it exhibits better integrity recognition of the entire target region. The experimental analysis demonstrates that our proposed method outperforms ASPP.

E. Comparison of the State-of-the-Art Methods

We compare the SRCBTFusion-Net with the other methods, including ABCNet [43] (based on Resnet50) and Deeplapv3+ [22] (based on Resnet50) based on pure CNN models, and TransUNet [10], Swin-Unet [44], UNetformer [45], and Segformer [46] transformer-based methods. Specifically, the design of TransUNet is similar to the UNet structure. The former four layers of resnet50 and transformer cascade are employed as dual encoders, and then the decoder is composed of upsampling and skip connection fusion. Swin-Unet designs the model of the UNet structure of a pure transformer. UNetformer adopts the UNet structure, in which Resnet50 is employed as an encoder, and three transformer stacked blocks and a CNN feature thinning head are employed as a decoder. Segformer adopts a transformer to obtain multiscale features and sends them to an MLP for segmentation.

To get the best segmentation effect, all the models we compare are trained with the pretraining weights supplied in their papers. In addition, the parameters and loss functions of the training models are the same as our models.

1) Results of the Potsdam Dataset: The test results of SRCBTFusion-Net and other methods are presented in Table IX. The segmentation results of SRCBTFusion-Net are better than other methods, with MIoU and Ave.$F1$ being 78.62% and 87.98%, respectively. Among them, Deeplabv3+ introduces the ASPP to capture the multiscale and large-scale features of deep semantic information and combines them with the bottom features to form the decoder. Although it improves the training speed, simple fusion will lead to insufficient fusion due to the gap between spatial position and semantic information. ABCNet utilizes a bilateral design to capture spatial context information and proposes a feature aggregation module to fuse two branches, but one branch only upsamples the initial input features and fuses the deepest features, ignoring the influence of middle-level features on the segmentation performance. Our SRCBTFusion-Net overcomes the defects of the former two models, and designs MFAM to process multiscale features with different dense connections,

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TABLE IX
COMPARISON OF RESULTS OF DIFFERENT METHODS ON THE POTSDAM (*Pot*) AND VAIHINGEN (*Vai*) DATASETS

Method	IoU (%)												Evaluation index			
	Impervious Surface		Building		Low Vegetation		Tree		Car		Clutter		MIoU (%)		Ave.F1 (%)	
	Pot	*Vai*	*Pot*	*Vai*	*Pot*	*Vai*	*Pot*	*Vai*	*Pot*	*Vai*	*Pot*	*Vai*	*Pot*	*Vai*	*Pot*	*Vai*
TransUNet [10]	79.21	80.72	86.12	87.21	69.87	61.88	70.44	73.55	78.66	68.07	30.48	**53.78**	76.86	74.30	86.76	84.95
ABCNet [43]	76.67	76.56	83.83	84.13	67.87	56.90	70.48	74.35	75.62	60.82	29.80	0.05	74.89	70.55	85.53	82.31
Deeplabv3+ [22]	78.18	79.40	85.69	86.47	**71.29**	59.73	72.36	76.20	79.02	71.72	29.22	34.33	77.31	74.70	87.11	85.21
Swin-Unet [44]	65.66	68.51	69.86	72.59	56.70	48.20	48.74	70.32	57.65	26.31	17.70	0.92	59.72	57.19	74.51	70.94
UNetformer [45]	79.10	79.76	**87.40**	86.63	70.29	60.02	71.68	76.61	80.18	71.73	30.34	4.14	77.73	74.95	87.32	85.38
Segformer [46]	78.45	80.69	86.15	87.47	71.01	61.30	**73.44**	77.03	78.65	69.63	**35.80**	6.61	77.54	75.23	87.24	85.55
SRCBTFusion-Net	**80.50**	**81.62**	87.23	**87.50**	71.17	**62.09**	72.52	**77.75**	**81.66**	**72.38**	33.15	0.38	**78.62**	**76.27**	**87.98**	**86.26**

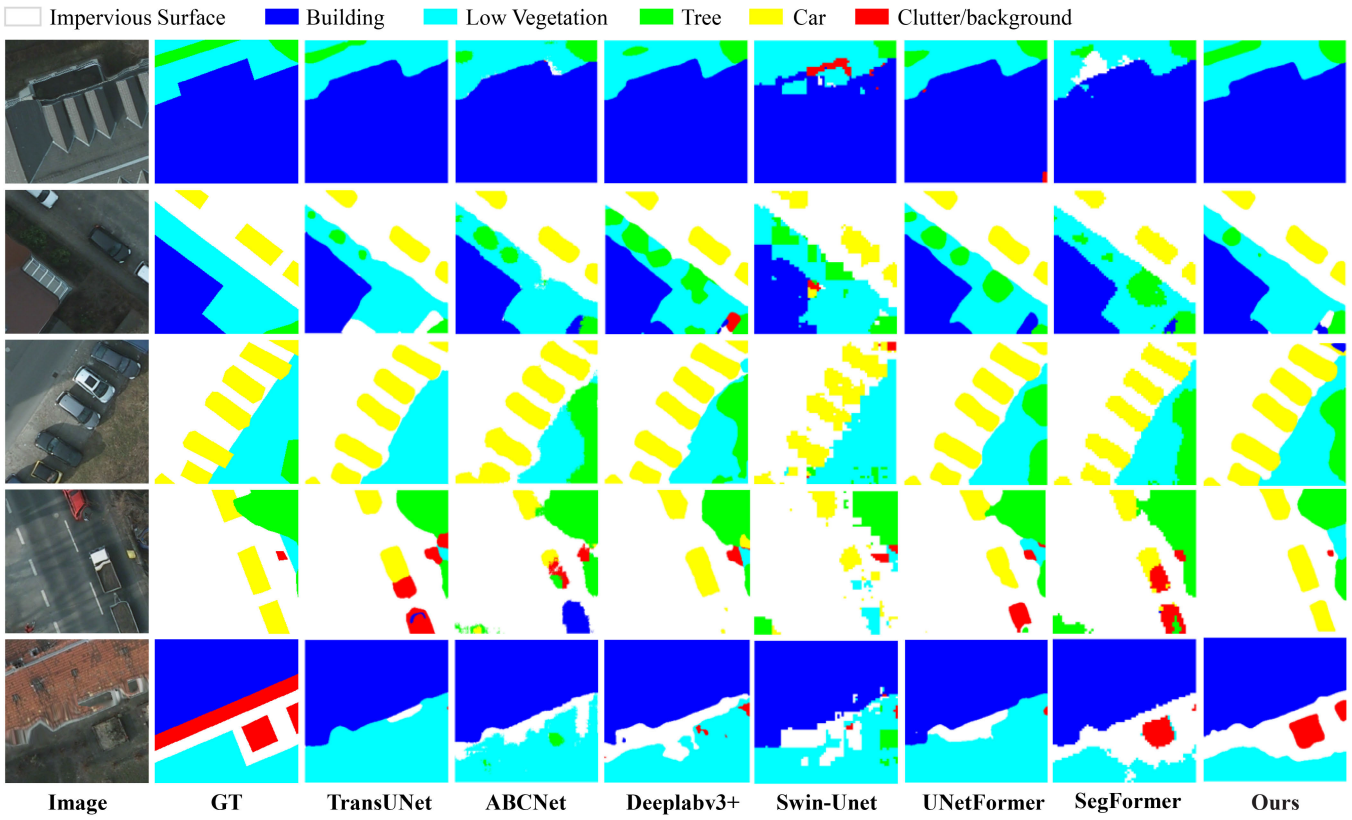


Fig. 13. Examples of semantic segmentation results of different models on the Potsdam dataset.

and then fuses them to get the same scale size and narrow the semantic gap information. Different from adopting the pure CNN model, Swin-Unet employs a pure transformer to build the UNet structure, which has advantages in obtaining global context information but ignores the importance of local information, so it does not perform well in segmentation. Segformer designs the structure of the transformer and MLP, which shows excellent segmentation effects. Unlike Segformer, UNetformer employs the UNet structure, which combines CNN and Transformer, and the segmentation result is second only to the proposed SRCBTFusion-Net. It is proved that the hybrid structure can effectively improve the segmentation performance. The MIoU of SRCBTFusion-Net achieves the most advanced results except for "LowVegetation" and "Tree," so our proposed network is more effective.

In Fig. 13, we visualize the results of several state-of-the-art models. Due to the lack of a combination of local and global information, Swin-Unet has not only fuzzy prediction in the semantic area but also has sawtooth texture on the boundary. Although Segformer has achieved good results in prediction accuracy, it only combines the composite structure of Transformer and MLP, without considering the advantages of CNN in extracting local information, and there is also apparent sawtooth texture at the edge. From the first, second, and fifth rows, we know that our proposed method can better segment the boundaries of "car" and "LowVegetation" and "low vegetation" and "Tree" which are easily confused. This is because our SR not only enhances contextual semantic information but also retains more spatial details. MASM captures rich low-level details, and precisely selects and integrates low-level features, thereby reducing the introduction of confusing information. The MFAM further makes up for the semantic difference between local and global, alleviates the loss of image feature information, and improves the ability to distinguish similar categories.

2) Results of the Vaihingen Dataset: To further prove the advantages of the SRCBTFusion-Net, we also compare the experimental results on the Vaihingen dataset in Table IX.

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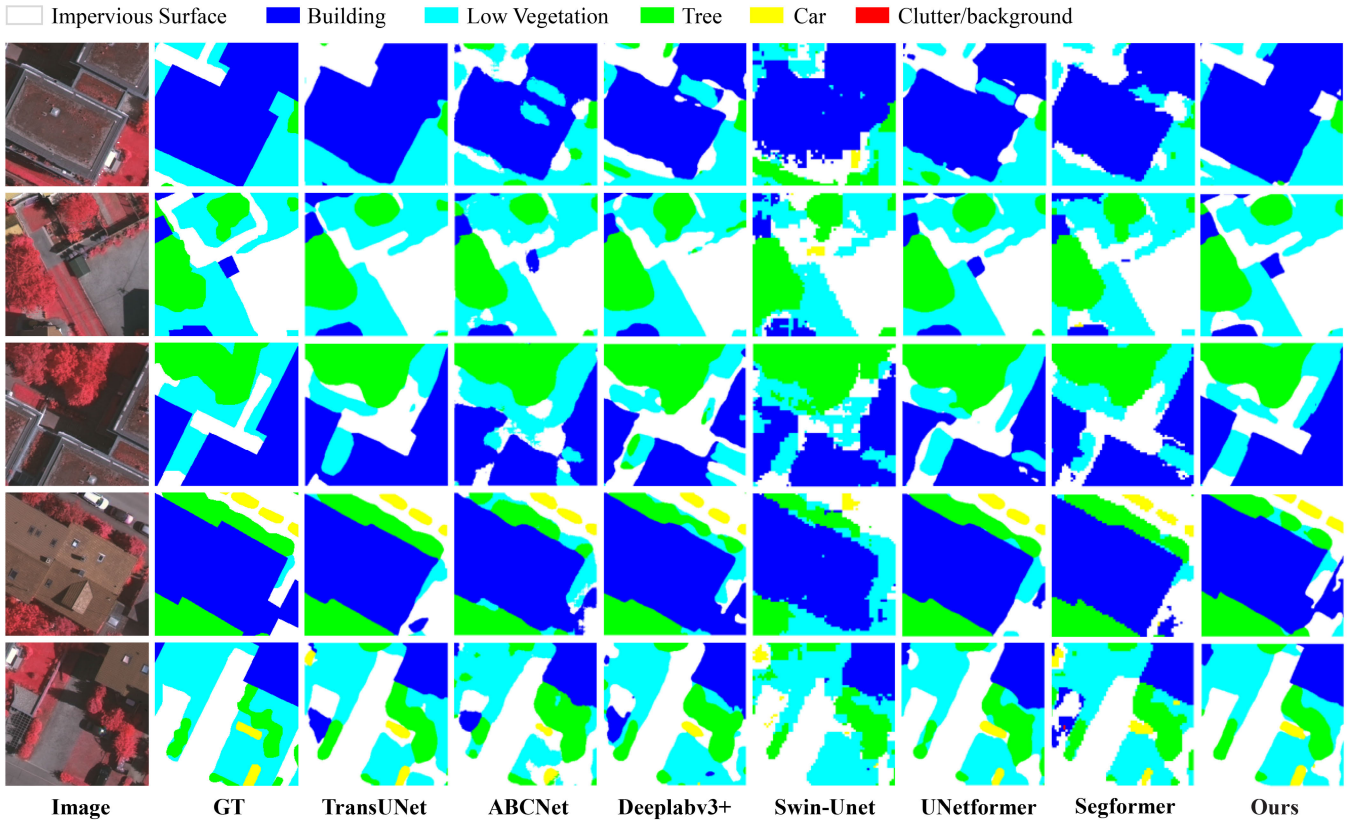


Fig. 14. Examples of semantic segmentation results of different models on the Vaihingen dataset.

SRCBTFusion-Net also achieves the best segmentation results, with MIoU and Ave.$F1$ being 76.27% and 86.26%, respectively. It is 1.04% and 0.71% higher than the best method Segformer. We can also find that the composite structure based on CNN and transformer is higher than the model based on pure CNN, which further shows that the composite structure is better than the pure CNN model by taking advantage of its own local and global modeling. Specifically, the IoU in the confusing "Impervious Surface" category is 0.90% higher than the best method TransUNet, which is due to the combination of SR and MFAM modules that we proposed to capture rich details and context information fully. In addition, SRCBTFusion-Net is improved based on TransUNet, and it is increased by 1.97% and 1.31% in MIoU and Ave.$F1$, respectively, which once again proves that our improvement is practical.

The visual results in Fig. 14 demonstrate the effectiveness of our approach. In the first row, shadow projections from the "Building" class onto the "Impervious Surface" class make segmentation extremely challenging for the latter. Many other methods struggle to accurately segment the "Impervious Surface" between the "Building" and "Low Vegetation" classes, but our method successfully accomplishes this task. In the second row, our approach cleanly segments complex regions among the "Building," "Impervious Surface," and "Low Vegetation" categories, providing strong evidence for the superiority of the proposed method. Currently, semantic segmentation models find it challenging to handle scenes with sharp corners and narrow pathways, as illustrated in the third and fourth rows. In these cases, many methods make segmentation errors, whereas our segmentation results approximate the true labels. In addition, in the fifth row, our method not only accurately distinguishes between "Building" and "Impervious Surface" but also successfully segments scenes with narrow pathways.

TABLE X
COMPARISON OF DIFFERENT PERFORMANCE METHODS

Method	Params (M)	Speed (FPS)	Flops (G)	Potsdam	Vaihingen
				MIoU (%)	MIoU (%)
TransUNet [10]	76.77	19	15.51	76.86	74.30
ABCNet [43]	28.57	26	7.24	74.89	70.55
Deeplabv3+ [22]	39.76	32	43.30	77.31	74.70
Swin-Unet [44]	41.42	22	0.02	59.72	57.19
UNetformer [45]	24.19	23	6.03	77.73	74.95
Segformer [46]	84.59	18	11.65	77.54	75.23
SRCBTFusion-Net	86.30	28	22.58	**78.62**	**76.27**

3) Complexity Comparison of Different Models: To compare the performance of the model comprehensively, the indicators that evaluate the performance of the model are tested in the same environment in Table X. Because our model adopts an eight-layer connected transformer as an encoder and multiscale feature fusion, the parameters are relatively large, but the FPS reaches 28, which exceeds TransUNet and Segformer with hybrid architecture. Because of the advantages of the pure CNN model, it is generally faster than the model based on a transformer in processing pictures. In addition, we also test the Flops (the smaller the model is, the less the computational complexity is). Although our computational complexity is slightly worse than other methods, the Flops of our SRCBTFusion-Net model exceed the representative pure CNN model Deeplabv3+. We also find that the Flops

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of Swin-Unet, a pure transformer model, are particularly small due to its use of the bidirectional encoder and decoder structures, avoiding cyclic structures such as recurrent neural network (RNN) so that the calculation of each position no longer depends on the calculation results of the preceding position. In addition, the feed-forward network in a transformer is based on a fully connected layer, which has parallel computing properties and can perform calculations quickly. Although the complexity of our model is not optimal, the best segmentation results are obtained on two datasets. Therefore, we explore whether the semantic segmentation hybrid model is meaningful.

V. Conclusion

To explore the impact of the hybrid architecture of SRCB and transformer on improving the segmentation results, and considering that the field of RS image segmentation has fuzzy boundary segmentation and lacks the ability to obtain enough context information, a dual coding network SRCBTFusion-Net combining SRCB and transformer is proposed. Specifically, SIEM and RGM are designed to enhance deep semantic information for improving the discrimination between edge features and nonedge pixel features. Through ablation experiments, it is proved that the proposed SRCBTFusion-Net is beneficial to better the smoothness of edge regions and upgrade the discrimination ability of different semantic regions. The design of the MASM enables more precise selection and integration of low-level features, thereby reducing the introduction of confusing information and further improving segmentation accuracy. In addition, MFAM is proposed to integrate multiscale features, enhance the extraction of semantic and contextual information, alleviate the loss of image feature information, and improve the ability to identify similar categories. Through experiments and analysis, it can be observed that the advantage of MFAM over FPN is that it better segments the blocked semantic areas, as well as the edges of the picture.

Although the proposed SRCBTFusion-Net has excellent advantages in the segmentation effect, it can not balance the parameters, operation speed, and segmentation accuracy well. In specific analysis, on the one hand, the cascade transformer improves the segmentation performance, and the number of parameters also increases considerably. On the other hand, the densely connected convolution blocks are utilized in MFAM, which decreases the operation speed of the model in a manner. Considering the foregoing concerns, we should focus on improving the structure of the transformer. In addition, the current method is challenging to segment the narrow-mixed area in the scene. In the future, we could benefit from some advanced ideas of road crack segmentation to further optimize the segmentation effect.

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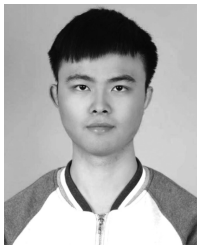
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